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Beyond the Joint: The Role of Central Nervous System Reorganizations in Chronic Musculoskeletal Disorders

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To a large extent, management of musculoskeletal disorders has traditionally focused on structural dysfunctions found within the musculoskeletal system, mainly around the affected joint. While a structural-dysfunction approach may be effective for musculoskeletal conditions in some populations, especially in acute presentations, its effectiveness remains limited in patients with recurrent or chronic musculoskeletal pain. Numerous studies have shown that the human central nervous system (CNS) can undergo plastic reorganizations following musculoskeletal disorders; however, they can be maladaptive and contribute to altered joint control and chronic pain. In this Viewpoint, we argue that to improve rehabilitation outcomes in patients with chronic musculoskeletal pain, a global view of the disorder that incorporates both central (neural) and peripheral (joint-level) changes is needed. We also discuss the challenge of evaluating and rehabilitating central changes and the need for large, high-level studies to evaluate approaches incorpo-

rating central and peripheral changes and emerging therapies.

Why Looking at Peripheral Structural Dysfunction May Not Be Sufficient in Chronic Presentations

A primary limitation of an approach that exclusively targets peripheral structural dysfunctions is its failure to recognize the complexity of the pain phenomenon, by focusing only on the nociceptive signal arising from peripheral receptors. As most clinicians have probably observed, the level of pain experienced by individuals with musculoskeletal disorders often varies substantially, even in cases where the structural failures within the joint are similar. This clinical observation is confirmed by several imaging studies that

have found weak association between symptoms related to musculoskeletal disorders and structural failure.⁴ Such disparities could be explained by central sensitization, a phenomenon defined as “an amplification of neural signalling within the central nervous system eliciting pain hypersensitivity” and considered a form of maladaptive plasticity.¹⁹ Indeed, patients with musculoskeletal disorders who express signs of central sensitization show higher pain severity and poorer treatment outcomes, as outlined in a recent Viewpoint paper that also addressed assessment and treatment.⁸

A second limitation of treatment approaches based on structural dysfunction is that they neglect the fact that the CNS can rapidly adapt to changes occurring at the periphery. For instance, the presence of acute pain, immobilization, or repeated use of new motor patterns—each a factor likely to be present following musculoskeletal disorders—has been shown to result in rapid alterations

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in sensorimotor cortex excitability. Brain reorganizations have been documented in populations with chronic musculoskeletal disorders when compared to healthy controls, both in the structural (eg, loss of gray matter) and functional (eg, changes in motor cortical representations) organization of the primary somatosensory and motor cortices (M1).

Assessment of motor cortical representations has mostly used transcranial magnetic stimulation applied over M1 to characterize changes in corticospinal excitability, intracortical inhibition/facilitation, and cortical maps. Changes in corticospinal excitability, which can reflect reorganization both at the cortical and spinal levels, have been documented in a number of musculoskeletal disorders, with some studies reporting increases in excitability (in lateral epicondylitis¹³ and low back pain [LBP]¹⁸) and others documenting reductions (rotator cuff tendinopathy⁷ and shoulder instability¹). At least a portion of these changes can be attributed directly to cortical changes, as alterations in intracortical inhibitory and facilitatory networks have also been documented in individuals with chronic musculoskeletal disorders.^{2,6} Measures of cortical maps give an estimation of the position and volume of the M1 representation of a given muscle. In individuals with lateral epicondylitis¹³ and with recurrent LBP,¹⁶ increased overlap of the spatial territory of the cortical representation of muscles of the painful body parts was observed. According to Schabrun et al,¹³ such overlap could lead to altered intermuscle coordination and loss of individuated movement that could limit the capacity to independently activate muscles during active movements and thus alter joint motor control. Furthermore, in individuals with recurrent LBP, changes in the spatial territory of the M1 representation of the transversus abdominis (TrA) were correlated with delayed onset of TrA electromyography during rapid arm movements (suggesting impaired anticipatory postural activation of trunk muscles),¹⁷ again suggest-

ing that changes in M1 organization are associated with altered motor control. Overall, these changes in cortical maps or in the balance between inhibition and excitation may disrupt the facilitation of the muscles needed to execute a motor task, as well as the inhibition of other muscles for fine tuning movement.² Such changes may predispose individuals to modify their motor strategies to control the joint, which could, in turn, contribute to chronicity through maladaptive plasticity. Persistence of M1 reorganizations upon return to pain-free movement execution might also favor the recurrence of symptoms.¹⁸ Cortical reorganization is not limited to motor representations, however, as expansion of cortical representation related to nociceptive input and increased cortical reactivity (sensory representations) have also been documented in musculoskeletal populations.³

Can Central Sensorimotor Changes Be Evaluated in the Clinic?

Central nervous system reorganization can occur relatively quickly. This rapid time course implies that injured individuals beginning rehabilitation may already have a "reorganized" CNS if they have experienced pain or prolonged limb immobilization, or have begun to move differently from normal to minimize pain. Therefore, when evaluating individuals with chronic or recurrent musculoskeletal disorders, clinicians should keep in mind that CNS reorganization may have already occurred. It can be complicated for clinicians to objectify the presence of such reorganizations using validated clinical tests, however, as current techniques are mostly laboratory based.

As previously mentioned, CNS reorganizations have been associated with changes in motor or postural control. One option for clinicians would therefore be to evaluate how the joint is controlled. Clinically, this can be assessed by observing how patients actively move their limb during a motor task, looking for changes in movement symmetry, appearance of compensations, or aberrant movements

(ie, scapular dyskinesis or shrugging during arm elevation, loss of proximal control during lower-limb squats, instability catch, painful arc of motion, or reversal of lumbopelvic rhythm during trunk flexion). One must be careful, however, as healthy controls have been shown to present with what can be viewed as altered motor control (eg, obvious scapular dyskinesis is present in 33% of healthy controls). Clinicians therefore have to determine whether the altered control is a contributing factor to their patients' condition. The presence of pain or painful arc of motion during altered movements may help determine whether it is significant. Pain relief when the altered movement is corrected (eg, by using manual feedback) might further confirm the significance of this alteration and help to guide treatment. Several standardized procedures have been developed, such as the Shoulder Symptom Modification Procedure, which looks at the influence of different maneuvers or corrections to alleviate symptoms during upper-limb movements.⁴

Clinical evaluation should also include the assessment of cognitive-emotional factors, such as fear avoidance, pain catastrophizing, and anxiety, as these factors can impact joint control (protection, avoidance) and central sensitization, and ultimately reinforce or consolidate maladaptive CNS reorganizations. Similarly, workplace and sport activities should be evaluated, as inadequate work stations or work/sport gestures might reinforce maladaptive motor strategies. In fact, these factors must all be considered during rehabilitation. **TABLE 1** presents elements that should be included in the clinical assessment when using a global approach.

How to Rehabilitate Central Sensorimotor Changes

It is possible to reverse maladaptive reorganizations, as the CNS has the capacity for adaptive plasticity. Studies have shown that changes in structural and functional brain organizations can be reversed in populations with chronic

TABLE 1

PRINCIPAL ELEMENTS OF THE CLINICAL ASSESSMENT OF CHRONIC MUSCULOSKELETAL DISORDERS

Structural-Dysfunction Approach

Structure(s) affected within the musculoskeletal system

- Orthopaedic diagnostic tests

Symptoms and functional limitations

- Self-reported questionnaires
- Clinical history taking and subjective examination

Posture

- Observation and objective measures

Mobility

- Active and passive range of motion
- Muscular and capsuloligamentous flexibility tests
- Joint mobility and stability (ie, gliding and translation)

Neurodynamic

- Neurodynamic tests

Strength and endurance

- Manual muscle testing, isometric muscle strength, or isokinetic peak torque
- Endurance tests

Biomechanics/motor control

- Qualitative observation of movements to determine whether they are within normality (ie, away from normal motor pattern)

Global Approach

All of the elements included in the structural-dysfunction approach, plus the following:

Motor control/biomechanics

- Qualitative observation of limb movement to determine movement alterations and association with symptoms
- Correction of altered movements and effect on pain

Fear avoidance, pain catastrophizing, and anxiety

- Self-reported questionnaires
- Fear of movement, avoidance, or protective reactions during the evaluation of mobility and motor control

Central sensitization

- Disproportionate pain experience
- Diffuse and widespread pain distribution
- Sensory hypersensitivity

Workplace and sport activities (or other leisure activities)

- Assessment of work stations or work/sport gestures

Other elements to consider

- Activity level
- Sleep
- Mood
- Stress management

musculoskeletal disorders following surgical (increased M1 cortical thickness¹⁵) or rehabilitation (normalization of M1 representations¹⁷) interventions. Central nervous system reorganization, therefore, should be one of the main targets during rehabilitation of chronic and recurrent musculoskeletal disorders, although interventions that do not specifically target the CNS, such as spine surgery or facet joint injections, have also been shown to lead to brain reorganizations.¹⁵ Still, rehabilitation protocols based on central adaptive plasticity are emerging and are showing promising results.^{5,12,14,17} In patients with musculoskeletal disorders, studies have documented that sensorimotor or skill training leads to improvements in motor performance and reductions in clinical symptoms. Several studies in populations with chronic LBP have also shown that programs involving sensorimotor training (ie, voluntary contractions of muscles to improve the control of the TrA, activation of deep multifidus muscles using real-time ultra-

sound imaging as feedback) lead to normalization of the M1 representations, as well as better postural control, improved functional level, and decreased symptoms.¹⁷ Using a more global approach that includes retraining an appropriate movement control could therefore be an approach to favor during rehabilitation in individuals with chronic conditions. Large, methodologically sound studies are, however, needed to prove that a global approach (as suggested in this Viewpoint) is more effective than a structural-dysfunction approach. To our knowledge, the two have never been directly compared.

We have been using sensorimotor training in individuals with rotator cuff tendinopathy,¹² with the objective of promoting better muscle coordination to centralize the humeral head and optimize scapular movement. This sensorimotor training is initially combined with modalities targeting peripheral structural dysfunctions (eg, stretching, strengthening, or mobilization) when peripheral

deficits are noted during the clinical evaluation (TABLE 2). Sensorimotor training starts with education on shoulder control and pain neuroscience, and on coaching about the rehabilitation program, to make sure that patients understand the goals and don't feel "threatened" by the exercises (TABLE 2). Thereafter, exercises of increasing difficulty in terms of movement plane, range of motion, speed, and resistance are performed during arm elevation movements. Feedback is provided when necessary by a physical therapist to elicit awareness of motor control deficits and ultimately improve shoulder control.¹¹ Progression is based on the level of resistance applied on the shoulder during arm elevation and on use or nonuse of feedback. Once patients reach pain-free execution and adequate control, the program is progressed into re-education exercises according to work, sport, and activities of daily living that they want to resume. Therefore, patients are supervised through graduated exposure to shoulder movements in elevated arm positions and movements often feared to be painful.⁹ The goal is to enhance intrinsic error detection to decrease the need for external help, thereby improving the patients' ability to better self-manage the program. While this program has been designed for the shoulder, a global view of motor rehabilitation suggests that similar steps can be used for other musculoskeletal disorders.

New therapies, such as noninvasive brain and peripheral neuromuscular stimulation, are also emerging to induce plasticity and to normalize CNS organization. They include transcranial direct-current stimulation (tDCS), shown to modulate neuronal membrane potential and neuronal excitability. Schabrun et al¹⁴ have looked at the short-term effect of tDCS (without any sensorimotor training) in individuals with recurrent LBP and found decreased pain severity, but no changes in M1 representations. However, when tDCS was combined with peripheral electrical stimulation, normalization of the motor representation was also

TABLE 2

PRINCIPAL ELEMENTS OF ACTIVE MANAGEMENT OF CHRONIC MUSCULOSKELETAL DISORDERS

Structural-Dysfunction Approach	Global Approach
Education on pain management, structures affected, and rehabilitation stages	All of the elements included in the structural dysfunction approach, except:
Postural exercises to realign limb/body segment	• Minimal use of passive techniques (if used, for goal-directed desensitizing)
Mobility exercises	• Normalization of posture and movement only if significant for the patient's painful condition
• Targeting muscular and capsuloligamentous structures	Education on motor control, pain neuroscience (ie, biopsychosocial pain mechanism), and graded exposure to exercise
Joint mobilization and manipulation	Sensorimotor training/graded exercises
Neurodynamic mobilization/desensitization maneuvers	• Using feedback to improve motor control by correcting movement alterations significant for patient's condition
Muscle strengthening exercises	• Emphasizing quality of movement control over quantity/speed/force
• Targeting strength and endurance	• Exercises of gradually increasing difficulty/graded activity
Exercises to normalize movement quality	• Work- or sport-specific re-education (goal-directed retraining)
• Normalization of movement	Building self-efficacy to self-manage symptoms
	• Education
	• Enhance intrinsic error-detection capabilities during sensorimotor training

observed.¹⁴ We have looked at the mid- and long-term effects of tDCS combined with sensorimotor training in individuals with rotator cuff tendinopathy and did not find any added effect when compared to sensorimotor training alone.¹⁰ The discrepancy between these results and those of Schabrun et al¹⁴ might be explained by the fact that sensorimotor training alone was very effective, potentially leading to a ceiling effect. Another technique showing some potential to regulate CNS organization and improve intervention effectiveness is repetitive peripheral magnetic stimulation applied over spinal nerve roots or over peripheral muscle. In one study of individuals with chronic LBP, the combination of repetitive peripheral magnetic stimulation and sensorimotor training was shown to be more effective in the short term than sensorimotor training alone to improve postural control and functional level, and to increase corticospinal excitability.⁵ While these studies have shown some promising results, more studies on larger cohorts and comparing these new emerging therapies to more traditional approaches are needed before implementing them in the clinic.

CONCLUSION

IN THIS VIEWPOINT, WE HAVE ARGUED that chronic musculoskeletal disorders may not be confined to the joint, as once thought, and that growing evidence shows that the maladaptive CNS reorganization associated with these disorders might be involved in altered joint control and chronic pain. As the integrity of the cortical networks is fundamental to normal joint control, it is necessary for clinicians to improve their understanding of the central neural mechanisms that lead to motor deficits and pain, and to tailor interventions that could optimize recovery or prevent chronicity and recurrence. ●

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